

Informatics II, Spring 2026, Solution 8

Binary Tree

Task 1

Task 1.1: correct: B,C Task 1.2: A Task 1.3: A Task 1.4: A

Task 2

See code in the file ex08.2.c.

Task 3

See code in the file ex08.3.c.

Task 4

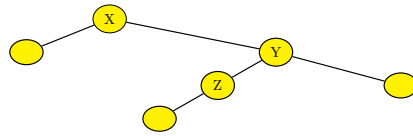
Task 4.1 Write a pseudocode algorithm `lftRot(X)` that left rotates node X.



Algorithm: `lftRot(X)`

```
Y = X → rgt;
Z = Y → rgt;
Y_left = Y → lft;
Y → lft = X;
X → rgt = Y_left;
X → hgt = 0;
Y → hgt = 1;
return Y
```

Task 4.2 Write a pseudocode algorithm **AVLrebalance(X, Y, Z)** that rebalances the following AVL tree.



Algorithm: AVLrebalance(X, Y, Z)

Z.right = Z → rgt;

Y → lft = Z.right;

Z → rgt = Y;

Y → hgt = 1;

Z → hgt = 2;

Z.left = Z → lft;

X → rgt = Z.left;

Z → lft = X;

Z → hgt = 2;

X → hgt = 1;

return Z

